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Assignments-1 (CO1 & CO5)

CO1: On completion of this course student will be able to apply fundamental concepts of chemistry in different fields of Engineering

Ques-1 What do you understand by liquid crystals? Discuss its classification & give their important application.

Ans- classification of LCs

1) orientational order
Molecules tend to align along a preferred direction called the director (Long Range order)

2) Positional order
Refers to the regular arrangement of molecules in space.

:) translation motion
:) short range order

Liquid Crystal (LCs)

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Liquid Crystal (L.Cs)

Thermotropic (L.Cs) (temp^{o} dependent)

Lytropic (L.Cs) (polar molecule)

- Nematic Liquid Crystal
- Smetic Liquid Crystal
- chlestic Liquid Crystal

Thermotropic Liquid Crystals.

These depend mainly on temp^{o} changes

a) Nomatic Liquid Crystals:

- Simplest & most common phase
- Molecule show orientational order only
- No positional order
- Easily aligned by electric or magnetic fields
- Used in LCD technology.

b) Smectic Liquid Crystals

- Passes both orientational and positional order.
- Molecules arranged in distinct layers.
- Layers can slide over each other.
- Found at lower temp than nematic phase

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9) Cholestoric Liquid Crystals.
- Also called chiral nematic phase
- Molecule arranged in helical structure
- Adjacent layer rotated at a fixed angle.
- Reflect specific wavelength of light

Lyotrophic liquid Crystals
These are formed by mixing two or more substance, where at least one is a polar molecule. Their formation depends on contraction and so went
Ex- Soap in water

Types include:
- Micellar cubic Phase
- Hexagonal phase.
- Lamellar phase

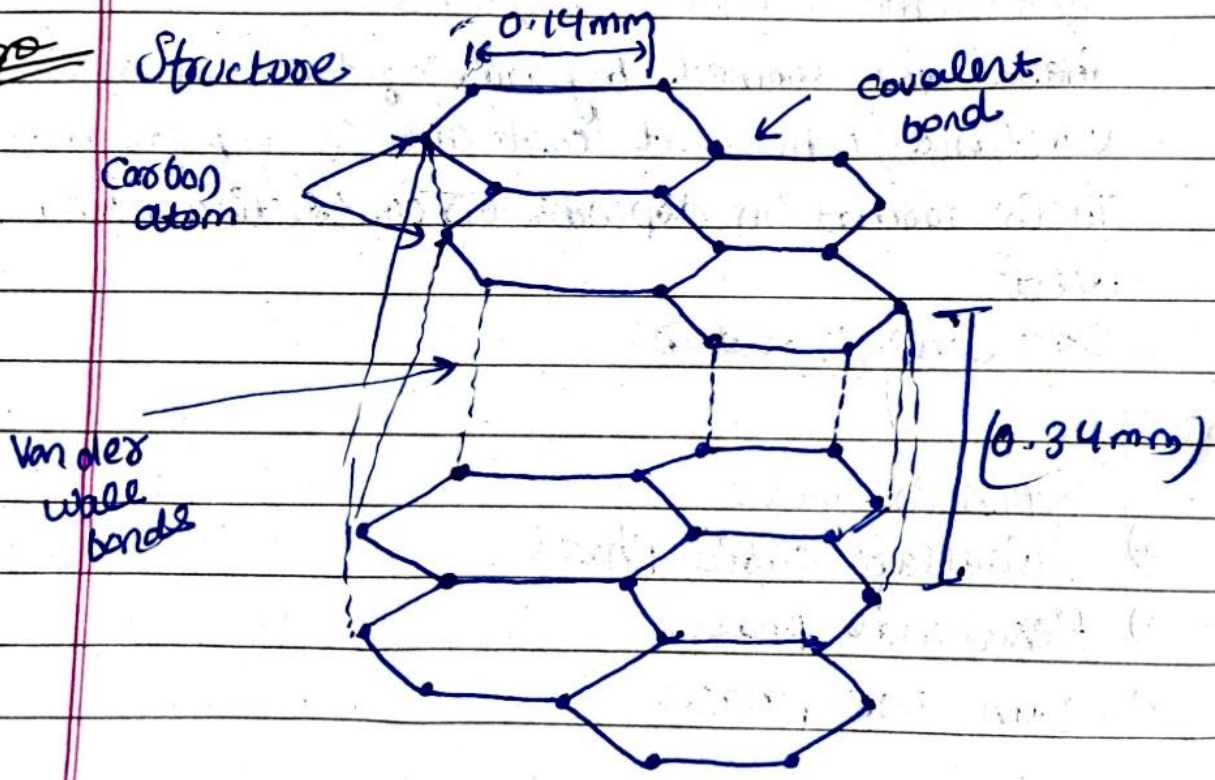
Application of liquid crystalls:
1) Liquid crystal Displays in mobiles, laptop, television.
2) Digital watches & Calculators
3) Thermometer & Temperature sensors

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4) Optical switches & modulators.
5) Medical diagnostic device

Q-2 Describe the structure & application of Graphite & fullerences. Explain the reasons for electrical & lubricating nature of Graphite?

Ans Structure



1) 'C's are sp^2 hybridised
2) Bond angle is 120°.
3) Each 'C' atom is attached to 3 neighbouring 'C' atom through strong covalent bonds

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4) 'C-C' bond length is 0.14 nm
5) Inter layer distance is 0.34 nm.
6) They are loosely bond through weak vanderwall force.

Lubricating & Electrical nature of Graphite:-

1) Due to the available loosely bound free e's Graphite shows electrical conductivity
2) Owing to weak vanderwall force in the graphite layer, the layers can easily slide on each other.
3) High thermal Stability

Applications

1) Refractory material owing to its high thermal stability & chemical inertness.
2) Anode material for electrochemistry process
3) Carbon brushes in electrical motor
4) Piston ring thrust bearing due to mechanical strength.

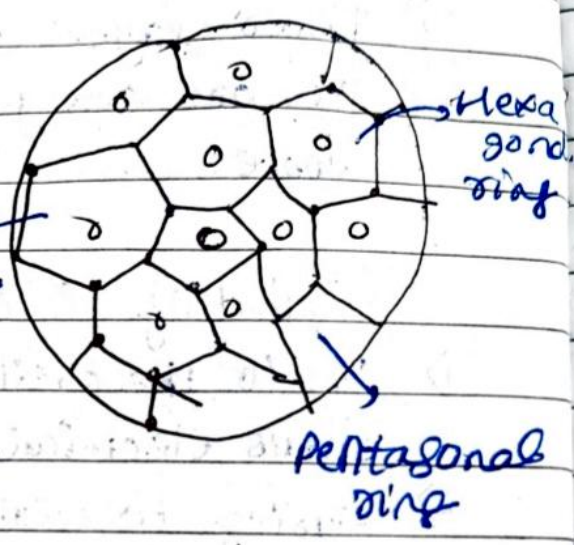
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Fullerence: Buck mister ball (bucky ball)

Structure:



1) 'C' are sp^2 hybridised
2) Consist of 20 Hexagonal 12 pentagonal rings.
3) 32 face & 60 Vertical.
4) No two pentagonal ring share a common atom.

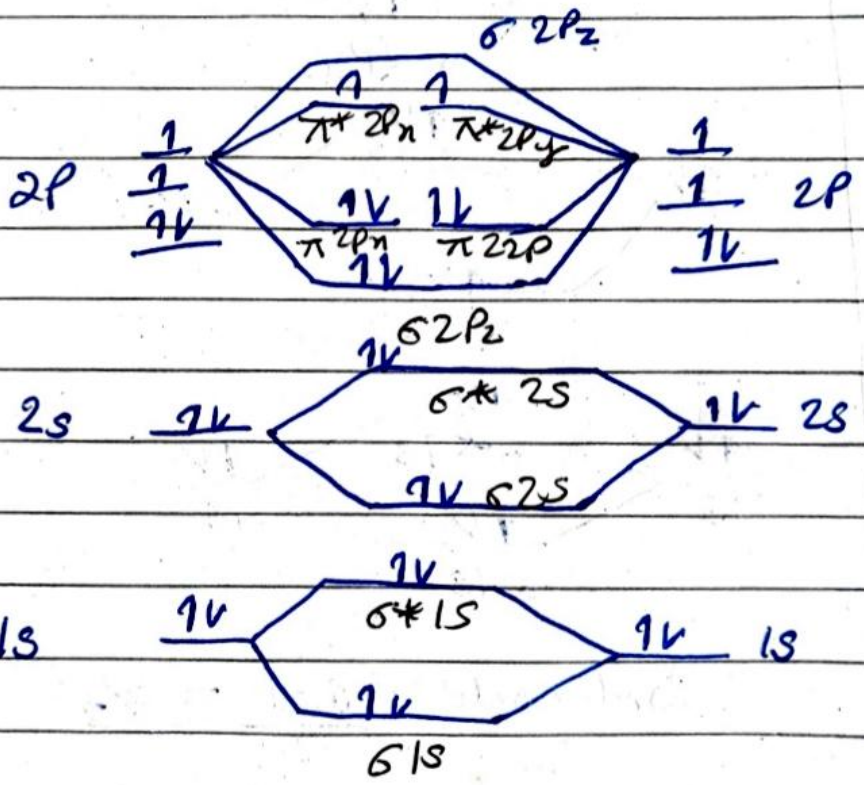
Application →
1) drug delivery agent and nanomedicines
2) As a lubricant
3) Catalyst
4) Reinforcing material

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Ques With the help of a molecular orbital diagram, explain the formation of O_2 & NO. Calculate B.O & product their magnetic behaviour.

Ans O_2



Bond order = $6-2/2 = 4/2 = 2$

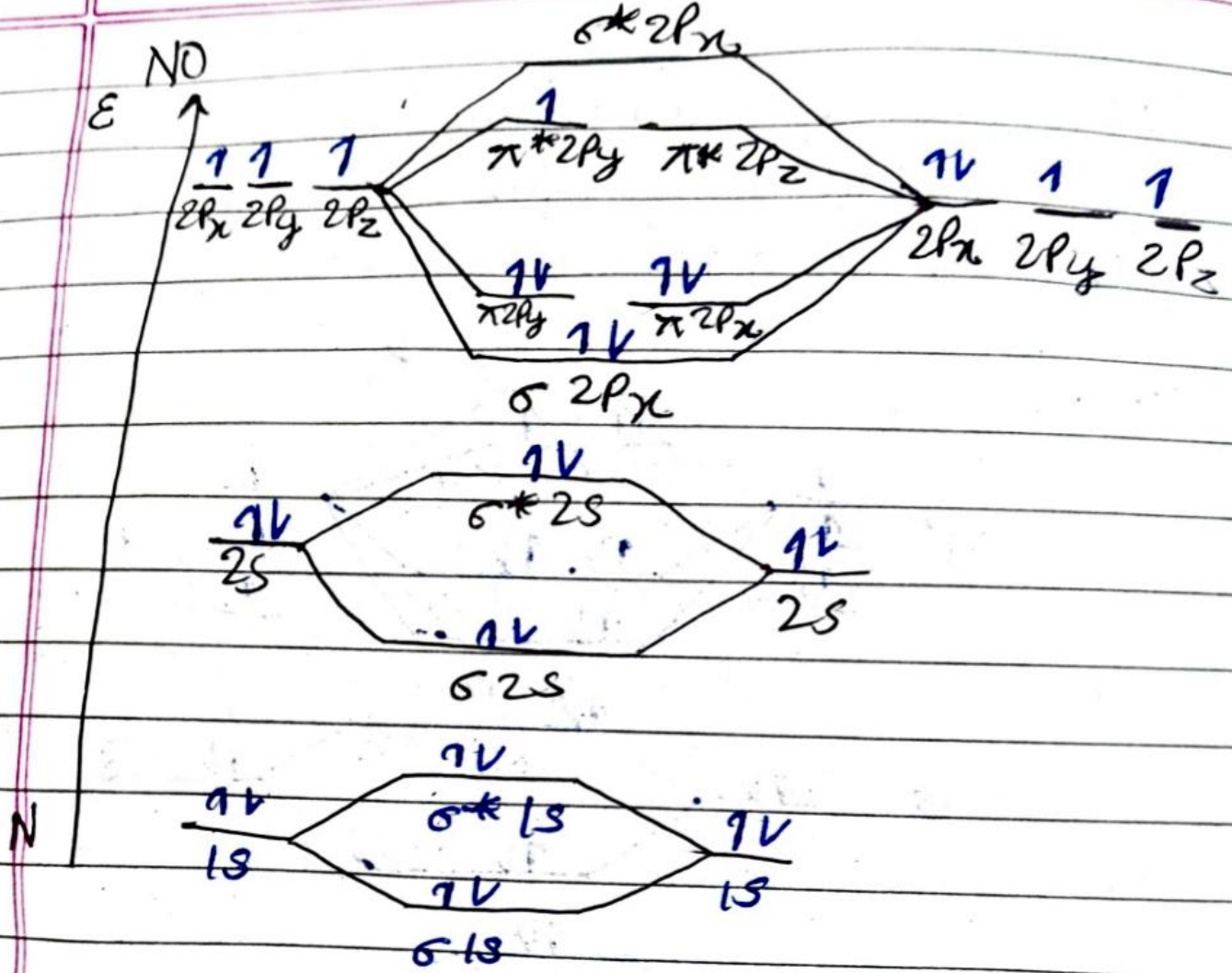
→ Paramagnetic

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Bond order = $\frac{6-1}{2} = 5/2 = 2.5$

Paramagnetic.